
LGCP simulation pipeline (rLGCP)
$i = 1, \ldots, N$
$N = n_{\text{train}} + n_{\text{val}} + n_{\text{test}}$
$\mu_i \sim \mathcal{U}(\mu_{\min},\ \mu_{\max})$
$\sigma_i^2 \sim \mathcal{U}(\sigma^2_{\min},\ \sigma^2_{\max})$
$s_i \sim \mathcal{U}(s_{\min},\ s_{\max})$
rLGCP
TARGETS
$\boldsymbol{\theta}_i = (\mu_i,\ \sigma_i^2,\ s_i)$
INPUTS
$L_c(r) = \hat{L}(r) - r$
N_i
$\mathbf{f}_i \in \mathbb{R}^8$
Quadrat-based (c_k)
$f_1 = \mathrm{Var}(c_k)$
$f_2 = \mathrm{Var}(c_k)/\bar{c}$
$f_3 = \max_k c_k / (\min_k c_k + 1)$
KDE-based $(\hat{\lambda})$
$f_4 = \mathrm{Var}(\hat{\lambda})$
$f_5 = \mathrm{skew}(\hat{\lambda})$
$f_6 = \mathrm{kurt}(\hat{\lambda})$
$f_7 = H(\hat{\lambda})$
$f_8 = \mathrm{SD}(\hat{\lambda})/\bar{\hat{\lambda}}$
Training
$i = 1, \ldots, n_{\text{train}}$
Validation
$i = 1, \ldots, n_{\text{val}}$
Test
$i = 1, \ldots, n_{\text{test}}$
Dataset splits